Unit - 3

MLT

Syllabus:-

Decision Tree Learning :- Decision tree learning algorithm, inductive bias, inductive inference with decision trees, Entropy and Information theory, information gain, ID-3 algorithm, Issues in Decision tree learning.

Instance-Based Learning :- K-nearest Neighbour learning, Locally weighted regression, radial basis function networks, case based learning.

Decision Tree Learning :-

Decision Tree Learning is a supervised machine learning algorithm used for classification and regression problems.

It works like a flowchart or a tree structure.

- **Root node** :- It represents the entire dataset.
- **Branches** :- represent decisions.
- **Leaf nodes** :- represent outcomes.

It's called a decision tree because it looks like a tree where you start at the root and follow branches.

Inductive Bias :-

Inductive Bias is the set of assumptions a learning algorithm makes to generalize from limited training data to unseen cases.

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- An algorithm learns from a limited set of training data.
- But training data is never enough to cover all possible ~~ou~~ situations.
- To make predictions on unseen data, the algorithm must assume something about the problem.

→ These assumptions are called inductive Bias.

Example :-

Imagine you have trained data :-

Input : [1 → output A],
[2 → output A],
[3 → output A].

Now, what about input 4 ?

- One algorithm might assume "outputs will stay the same → A".
- Another might assume "pattern will change → maybe B".

Both fit the training data, but the inductive bias decides which prediction the algorithm makes for unseen cases.

Inductive Inference :-

Inductive inference simply means :-
Learning from examples and making rules to use on new situations.

In decision Trees :-

1.) We give examples (training data) :-
- Sunny, Hot → Don't play

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- Overcast, cool → play
- Rainy, mild → play.

2.) The tree learns rules from these examples:-
- If weather = overcast → play.
- If weather = sunny → don't play.
- If weather = rainy → play.

3.) Now we get a new case (not in training):
- Weather = sunny, mild.
- The tree uses the learned rule → predict don't play.

The "guessing from past to new" is inductive inference.

Entropy :-

⊙ Entropy is a measure of impurity or uncertainty in data.

⊙ In decision tree learning, it tells us how mixed the classes are in a dataset.

→ If data is pure (all ex. belong to one class).

$$\boxed{\text{Entropy} = 0}$$

→ If data is perfectly mixed (50% yes, 50% no)

$$\boxed{\text{Entropy} = 1.}$$

formula :-

$$\boxed{\text{Entropy}(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)} \quad ***$$

where :-

p_i = probability of class i.

c = number of classes

S = Total no. of samples.

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Example :-

Suppose we have 8 training sets.

Red fruits = 2

Green fruits = 2

Yellow fruits = 4.

$(P_i) \log_2 P_i$

Step 1 :-

$$P(\text{Red}) = \frac{2}{8} = \frac{1}{4}$$

$$P(\text{Green}) = \frac{2}{8} = \frac{1}{4}$$

$$P(\text{Yellow}) = \frac{4}{8} = \frac{1}{2}.$$

Step 2 :- Put into the entropy formula :-

$$E = -\left(\frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{1}{2}\log_2\left(\frac{1}{2}\right)\right)$$

$$E = -\left(\frac{1}{4} * (-2) + \frac{1}{4}(-2) + \frac{1}{2} * (-1)\right)$$

$$E = -\left(-\frac{1}{2} - \frac{1}{2} - \frac{1}{2}\right)$$

$$E = -(-1.5)$$

$$\boxed{E = 1.5}$$ Ans

Information Theory

- Entropy
- Information Gain

⊙ Information Theory is the study of measuring uncertainty and information.

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Information Gain :-

- Information Gain is a measure used in Decision Tree Learning.
- It tells us how much "information" a feature gives us about the class.
- In simple words :-

It shows how well a particular attribute splits the data into correct places.

$$\boxed{IG(S, A) = Entropy(S) - \sum_{v \in Values} \frac{|S_v|}{|S|} \times Entropy(S_v)}$$

where :-

S = dataset
A = attribute

Ex :- Suppose we are classifying whether a student passes or fails based on "study hours".

Pass →	6
Fail →	4

$\frac{6}{10}$

$$Entropy(S) = -\left[\frac{6}{10}\log_2\frac{6}{10} + \frac{4}{10}\log_2\frac{4}{10}\right]$$

$$\approx 0.97$$

Now, split by attribute "study hours > 5".

Group 1 (Yes) :- 5 students → 4 pass, 1 fail
Entropy ≈ 0.72

Group 2 (No) :- 5 students → 2 Pass, 3 fail
Entropy ≈ 0.97.

Weighted entropy after split :-

$$\frac{5}{10}(0.72) + \frac{5}{10}(0.97) = \underline{0.845}$$

So, information gain :-

$$\boxed{\begin{aligned} IG &= 0.97 - 0.845 \\ &= 0.125 \text{ Ans.} \end{aligned}}$$

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ID3 algorithm :-

- ID3 is an algorithm used to build Decision Trees.
- It uses information gain and entropy.

Steps :- (algorithm) :-

1.) Calculate the entropy of the dataset.
2.) For each attribute calculate information Gain.
3.) Choose the attribute with the highest information gain.
4.) Repeat the process for each branch until:
   a) all samples are of the same class.
   b) no attributes are left.

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	~~No~~ Yes
D6	Rain	Cool	Normal	Strong	~~Yes~~ No
D7	Overcast	Cool	Normal	Strong	~~No~~ Yes
D8	Sunny	mild	High	Weak	~~Yes~~ No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	mild	High	Strong	No

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Step 1 :-

Calculate the entropy :- (for each attribute).

Attribute (outlook) :- (values :- sunny, overcast, Rain).

for sunny :-

Yes = 2
No = 3

$$S_{sunny} = -\left[\frac{2}{5}\log_2\frac{2}{5} + \frac{3}{5}\log_2\frac{3}{5}\right]$$

$$= -\left[0.4\log_2 0.4 + 0.6\log_2 0.6\right]$$

$$= 0.47$$

For overcast :-

Yes = 4
No = 0

$$S_{overcast} = -\left[\frac{4}{4}\log_2\frac{4}{4} + 0\right]$$

$$= -\left[1\log_2 1 + 0\right]$$

$$= 0.$$

for rain :-

Yes = 3
No = 2

$$S_{rain} = -\left[\frac{3}{5}\log_2\frac{3}{5} + \frac{2}{5}\log_2\frac{2}{5}\right]$$

$$= -\left[0.6\log_2 0.6 + 0.4\log_2 0.4\right]$$

$$= 0.971$$

Yes = 9
No = 5

$$S_{Total} = -\left[\frac{9}{14}\log\frac{9}{14} + \frac{5}{14}\log_2\frac{5}{14}\right]$$

$$= 0.94$$

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Gain (S, outlook) :-

$$\Rightarrow \text{Entropy}(S) - \sum_{v \in (\text{sunny, overcast, rain})} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\Rightarrow 0.94 - \frac{5}{14} \text{Entropy}(\text{sunny}) - \frac{4}{14} \text{Entropy}(\text{overca}$$

$$- \frac{5}{14} \text{Entropy}(S_{rain}).$$

$$\Rightarrow 0.94 - \frac{5}{14} \times 0.971 - \frac{4}{14} \times 0 - \frac{5}{14} \times 0.971$$

$$\Rightarrow 0.2464$$

$$\boxed{\text{Gain}(S, \text{outlook}) = 0.2464}$$ ***

Attribute (Temperature) :- values – Hot, cool, mild

$S_{hot} \Rightarrow$ Yes = 2, No = 2 $\Rightarrow S_{mild} \Rightarrow$ Yes = 4, No = 2

$$S_{hot} = -\left[\frac{2}{4} \log_2 \frac{2}{4} + \frac{2}{4} \log_2 \frac{2}{4}\right] = 1.0$$

$$S_{mild} = -\left[\frac{4}{6} \log_2 \frac{4}{6} + \frac{2}{6} \log_2 \frac{2}{6}\right] = 0.9183$$

$$S_{cool} = -\left[\frac{3}{4} \log_2 \frac{3}{4} + \frac{1}{4} \log_2 \frac{1}{4}\right] = 0.8113$$

$$S_{Total} = -\left[\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14}\right] = 0.94.$$

Gain :-

$$\text{Gain}(S, \text{Temp}) = \text{Entropy}(S) - \sum_{v \in \text{hot, mild, cool}} \frac{|S_v|}{|S|} \text{Entrop}(S_v)$$

$$= 0.94 - \frac{4}{14} \times 1.0 - \frac{6}{14} \times 0.9183 - \frac{4}{14} \times 0.81$$

$$\boxed{\text{Gain}(S, \text{Temp}) = 0.0289}$$

↳ Repeat this process for other two attributes

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Attribute (humidity) :-

$\boxed{\text{Gain} = 0.1516}$

Attribute (wind) :-

$\boxed{\text{Gain} = 0.0478}$

The maximum value is the root node.
So,

$\boxed{\text{outlook}}$ → (root node)

Instance Based Learning :-

- Instance-Based Learning is a type of learning where the algorithm does not build an explicit model.
- Instead, it stores training examples (instances) and uses them directly to make predictions for new data.
- That's why it's often called "lazy learning" → because no model is built in advance.
- Some of the popular instance-based learning algorithms are :-

1) KNN → most common.
2.) Locally weighted regression.
3.) Case-Based learning.

1.) KNN :-

K-NN (K-nearest neighbours) is a type of instance-based learning.

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- It can also be used for regression.

Algorithm :-

1.) Choose K (the no. of neighbors to consider).

2.) Calculate distance b/w the new data point and all training points.

3.) Pick K nearest neighbors (the closest points).

4.) Classify :-

- If classification → take majority vote.
- If regression → take the average value.

Ex :-

Apply KNN for the following dataset and predict class of data example.

$(A_1 = 3, A_2 = 7)$

(Assume $K = 3$)

A_1	A_2	Class
7	7	True
7	4	True
3	4	False
1	4	True
5	3	False
6	3	True

$$d_1 = \sqrt{(x_1 - a)^2 + (y_1 - b)^2}$$

$$d_1 = \sqrt{(3-7)^2 + (7-7)^2}$$

$$\boxed{d_1 = 4}$$

$$d_2 = \sqrt{(3-7)^2 + (7-4)^2}$$

$$\boxed{d_2 = 5}$$

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$d_3 = \sqrt{(3-3)^2 + (7-4)^2}$

$\boxed{d_3 = 3}$

$d_4 = \sqrt{(3-1)^2 + (7-4)^2}$

$d_4 = \sqrt{13}$

$\boxed{d_4 = 3.6}$

$d_5 = \sqrt{(3-5)^2 + (7-3)^2}$

$d_5 = \sqrt{20}$

$\boxed{d_5 = 4.47}$

$d_6 = \sqrt{(3-6)^2 + (7-3)^2}$

$\boxed{d_6 = 5}$

(II) Arrange in ascending order.

3, 3.6, 4, 4.7, 5, 5

d_3, d_4, d_1, d_5, d_2, d_6.

(III) $k = 3$, observe 1st 3 classes.

$\boxed{T, F, T}$

A1	A2	Class
7	7	T
7	4	T
3	4	F
1	4	T
4	3	F
6	3	T

← d_1

← d_3

← d_4

→ True

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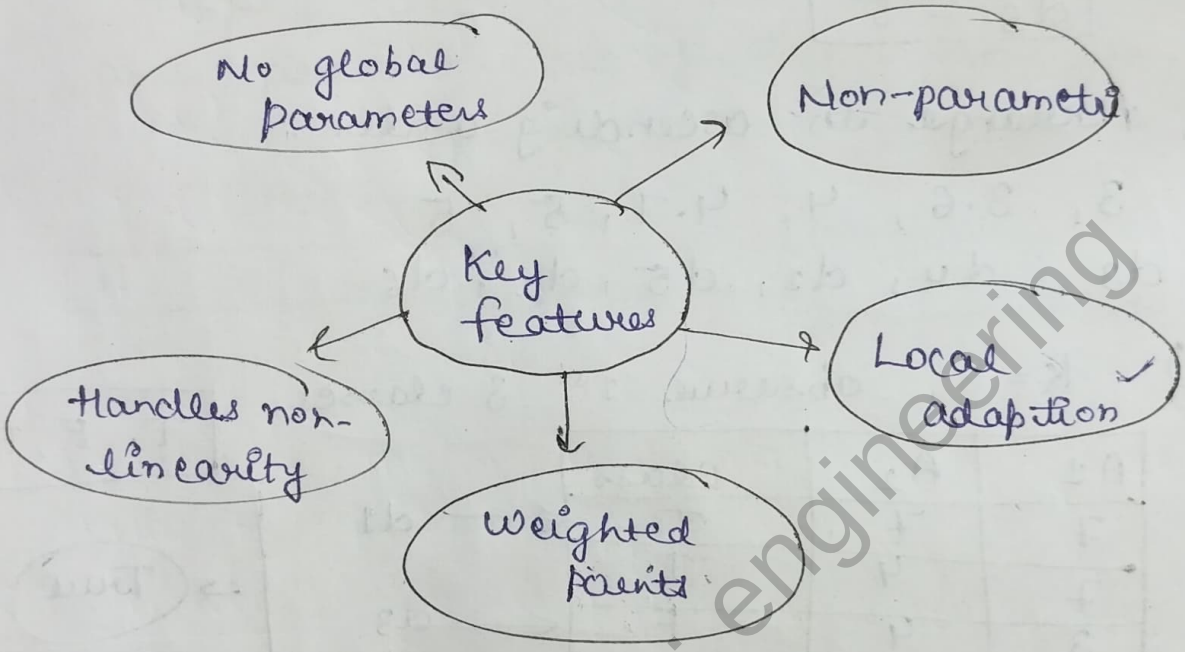
2.) Locally weighted Regression :-

- It is a type of instance based learning.
- Training points near the query point get higher weight.
- Farther points get lower weight.
- It is a memory based method that performs a regression around a point.
- It is a supervised learning algorithm.
- It is the generalization of K-nearest neighbour.

Key features :-



1.) Non-parametric :-
No global model, computes unique parameters for each query point.

2.) Local adaption :-
focuses on nearby data points for predictions.

3.) Weighted points :-
Closer points have more influence, distant points have less

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3.) Case Based Learning :-

- Case-Based Learning is a type of instance-based learning.
- Instead of building a general model, it stores past cases.
- When a new problem comes, it :-
  1.) find similar past cases.
  2.) Uses their solutions to solve the new problem.
  3.) Sometimes adapts the old solution if needed.
- The process typically involves four main steps :- retrieve, reuse, revise, and retain (4R's)

a) **Retrieve** :- It is the first step. The goal is to identify cases that are so close to the current problem as possible.

b) **Reuse** :- Once relevant cases have been retrieved, the next step is to reuse them to solve the current problem.

c) **Revise** :- After a sol^n has been developed using past cases, the next step is to revise it to better fit the current problem.

d) **Retain** :- It is the final step. Adding the new case to the case library so that it can be used in the retrieval step for future problem.

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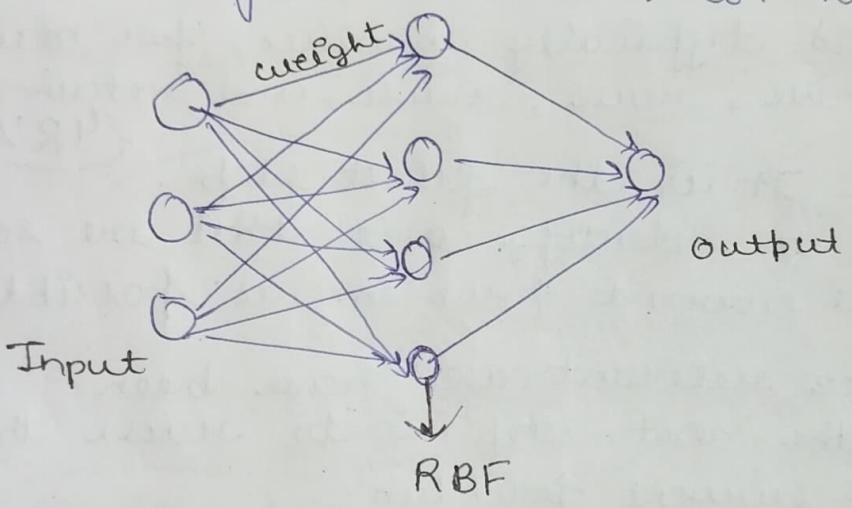
Radial Basis Function Network :-

- A radial basis function network is a type of artificial neural network.
- It is mainly used for classification, regression and function.

In short :-

Radial Basis function network :- A neural network that uses radial basis functions (like gaussian curves) in hidden neurons.



1.) **Input** :- The input layer simply feeds the data to hidden layer.

2.) **Hidden** :- It is based on Cover's theorem of patterns

3.) **Output** :- The output layer uses a linear activation function for both classification & regression.

$$\boxed{y = \sum_{i=1}^{n} W_i \phi_i} \quad ***$$

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Issues in Decision Tree Learning

The main issues in decision tree learning are :-

1.) **Overfitting :-**

A tree may become too complex and fit the training data perfectly.

Solution :- Use pruning, limit tree graph.

(2 marks) Pruning is a process of deleting the unnecessary nodes from a tree.

2.) **Missing values :-**

Real-world datasets often have missing attribute values, and its hard to decide which branch to follow.

3.) **Noisy data :-**

Noisy data is the data that has errors, mistakes.

4.) Bias toward attributes with many values.

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